

ECS_TASK

Task:

- ❖ Setup a highly available ecs cluster with load balancer and dynamic port mapping.

NOTE:

Your cluster should maintain atleast 6 tasks at any point of time and should be highly availble across multiple AZ's.

Use the below image from deployment:

[sabair0509/hiring-app:works](#)

PHASE 1: Create ECS Cluster (EC2 + ASG)

1 On this screen set:

- Provisioning: **On-demand**
- Instance type: **t3.medium** (recommended)
- Instance role: **Create default role**
- Desired capacity: **2**
- Minimum: **2**
- Maximum: **4**
- Select **2 subnets in different AZ**
- Select key pair

The screenshot shows the 'Create cluster' page in the AWS Management Console for Amazon Elastic Container Service. The page is titled 'Cluster configuration' and includes the following sections:

- Cluster name:** A text input field containing 'hiring-cluster'. Below it, a note states: 'Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).'.
- Service Connect defaults - optional:** A section for configuring service defaults.
- Default namespace:** A text input field with a search icon and a 'Create a new namespace' button.
- Infrastructure - advanced:** A section for configuring infrastructure, including a note: 'Configure the manner of obtaining compute resources that will be used to host your application.'
- Select a method of obtaining compute capacity:** A section with three radio button options:
 - Fargate only:** 'Serverless - you don't think about creating or managing servers. Great for most common workloads.'
 - Fargate and Managed Instances:** 'Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.'
 - Fargate and Self-managed instances:** 'Self-managed instances - you must ensure the instances are patched and scaled properly, and you have full control over the instances.'
- Auto Scaling group (ASG):** A section with two radio button options:
 - Create a new Auto Scaling group - advanced:** (Selected)
 - Use an existing Auto Scaling group**
- Provisioning model:** A section with two radio button options:
 - On-demand:** 'With on-demand instances, you pay for compute capacity by the hour, with no long-term commitments or upfront payments.' (Selected)
 - Spot:** 'Amazon EC2 Spot instances let you take advantage of unused EC2 capacity in the AWS cloud. Learn more'

AWS
[Alt+?] Search
United States [Bz, Virginia] ec2

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Tell us what you think

Create cluster

Subnets

Select the subnets where your instances are launched and your tasks run. We recommend that you use three subnets for production.

Clear current selection

subnet-045c03ac3c3f9b61
us-east-1a 172.31.80.0/20

subnet-0c3ff99509bc7fa6a
us-east-1a 172.31.80.0/20

Security group

[info](#)

☒ Create a new security group

☐ Use an existing security group

Security group name

Choose an existing security group:

Choose security groups

default

Auto-assign public IP

[info](#)

Choose whether to auto-assign a public IP to the Amazon EC2 instances

Turn off

▼ Monitoring - optional

Configure observability, encryption, and logging options to maintain compliance and operational visibility of your container environment.

Container insights

[info](#)

CloudWatch Container Insights is a monitoring and troubleshooting solution for containerized applications and microservices.

Select the level of observability you want to achieve with Container Insights

Container Insights is turned off by default in your Account Settings. You can enable that here at the cluster level.

- ☐ **Container Insights with enhanced observability** [info](#)
- ☐ Provides detailed health and performance metrics at task and container level in addition to aggregated metrics at cluster and service level. Enables easier drill-downs for faster problem isolation and troubleshooting.
- ☐ **Container Insights**
- ☐ Provides aggregated metrics at cluster and service level. You can run deep dive analysis with Logs Insights analytics.
- ☒ **Turned off**
- ☐ Provides default CloudWatch metrics only.

ECS Exec encryption and logging

[info](#)

 **Create cluster**

Wait 5-10 mins.

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Cluster hiring-cluster creation is in progress.

Cluster hiring-cluster

ARN

arn:aws:ecs:us-east-1:060795932818:cluster/hiring-cluster

Status

Active

CloudWatch monitoring

Default

Registered container instances

1

Services

Active

1

Tasks

Pending

1

Running

1

Services

Tasks

Infrastructure

Metrics

Scheduled tasks

Configuration

Event history

Tags

Services (0)

Filter services by value

Filter launch type

Any launch type

Filter scheduling strategy

Any scheduling strategy

Filter resource management type

Any resource management type

Service name

ARN

Status

Scheduling strategy

Launch type

Task definition

Deployments and tasks

Last deployment

Created at

No services

No services to display.

Create

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

Docker_controller

i-028f1dc5ea91e1b889

Stopped

t2.large

View alarms +

us-east-1c

Jenkins-Server

i-0ee33a1308b0a3bcb

Stopped

t3.large

View alarms +

us-east-1c

ECS Instance - hiring-cluster

i-0c4177b4f3bca229

Running

t3.medium

Initializing

View alarms +

us-east-1b

Master_jenkins

i-04e6d922f331d1cf

Stopped

t2.large

View alarms +

us-east-1a

MyAbleInstance

i-005f4e90166464efa

Stopped

t2.micro

View alarms +

us-east-1a

ansible-master

i-0161f3b5f92425c4

Stopped

t2.micro

View alarms +

us-east-1a

worker-1

i-0573f56e2eb008b7e

Stopped

t2.micro

View alarms +

us-east-1a

worker-2

i-09a0572904ab81bdc

Stopped

t2.micro

View alarms +

us-east-1a

ECS Instance - hiring-cluster

i-0e7fa246219b20fc2

Running

t3.medium

2/2 checks passed

View alarms +

us-east-1a

docker-ansible

i-009912a531a00b3f4

Stopped

t2.large

View alarms +

us-east-1c

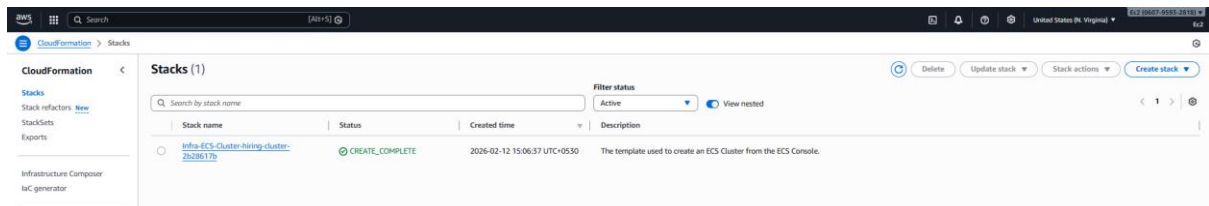
✅ PHASE 2: Confirm EC2 Instances Registered in ECS

Go to:

ECS → Clusters → hiring-cluster → Infrastructure

Check:

- EC2 instances should show as **ACTIVE**
- Container instances should show registered



If it shows 0 instances, wait 2-3 mins.

Yes bro. From Step 2 onward (cluster already creating), do this exactly:

✅ STEP 2: Verify EC2 Instances Joined ECS Cluster

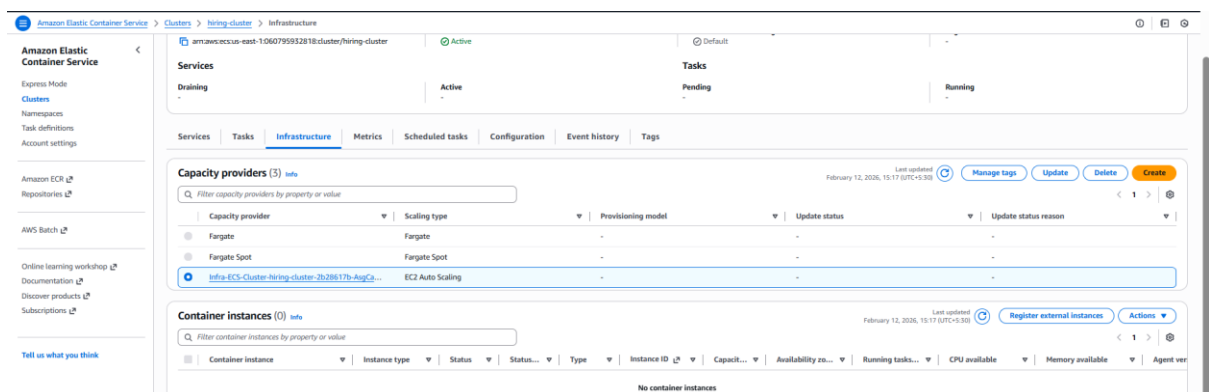
Go to:

ECS → Clusters → your-cluster → Infrastructure

Check:

- Container instances = 2 (or more)
- Status = ACTIVE

If 0, wait 2-3 mins.



✓ STEP 3: Create Application Load Balancer (ALB)

Go to:

EC2 → Load Balancers → Create Load Balancer → Application Load Balancer

Settings:

- Name: hiring-alb
- Scheme: Internet-facing
- Listener: HTTP 80
- VPC: Default VPC
- Subnets: select 2 subnets (2 AZ)

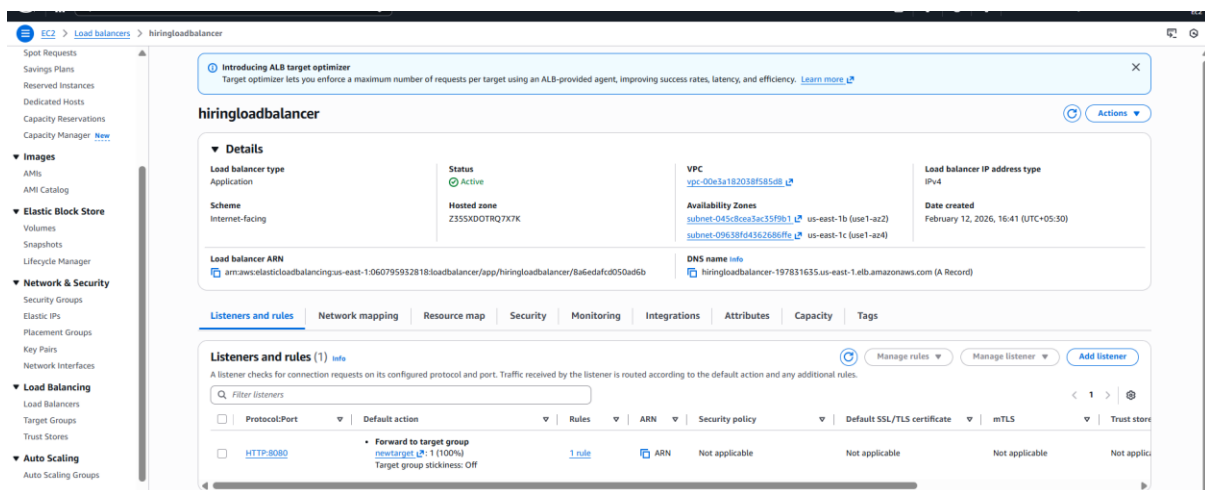
Create Security Group for ALB:

Name: alb-sg

Inbound rule:

- HTTP 80 from 0.0.0.0/0

Create ALB.



✓ STEP 4: Create Target Group

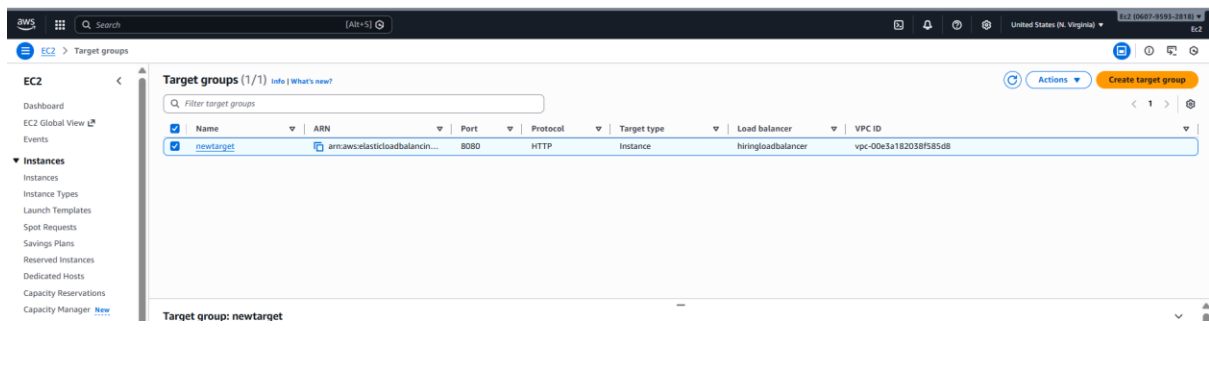
Go to:

EC2 → Target Groups → Create target group

Settings:

- Target type: ☒ Instance
- Protocol: HTTP
- Port: 80
- Name: hiring-tg
- Health check path: /

Create target group.



☒ STEP 5: Attach Target Group to ALB Listener

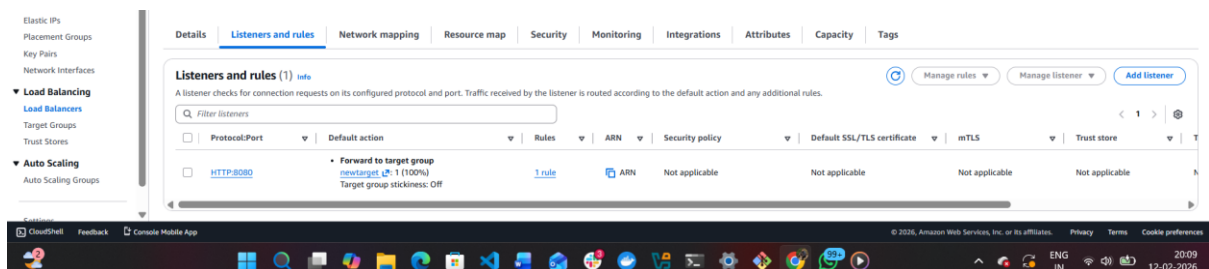
Go to:

EC2 → Load Balancers → EC2loadbalancer → Listeners → HTTP:80

Edit rule → Forward to:

☒ hiring-tg

Save.



☒ STEP 6: Update ECS EC2 Security Group (Very Important)

Go to:

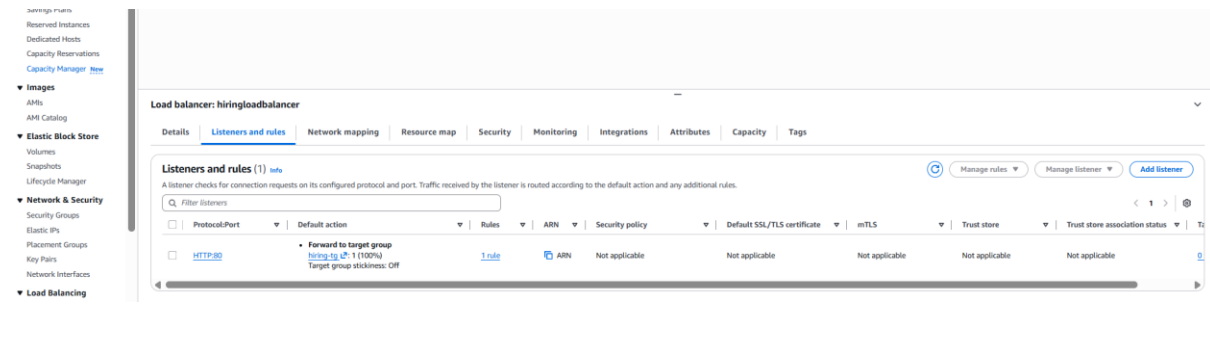
EC2 → Instances → select ECS instance → Security → open Security Group

Edit inbound rules and add:

Allow from ALB SG:

- HTTP 80 → source: alb-sg
- TCP 32768-65535 → source: alb-sg

(These ports are needed for dynamic port mapping)



✓ STEP 7: Create Task Definition (Using Given DockerHub Image)

Go to:

ECS → Task Definitions → Create new task definition

Choose:

✓ EC2

Task definition name:

hiring-task

Container:

Container name: hiring-app

Image:

sabair0509/hiring-app:works

Port mapping:

- Container port: 8080
- Host port: ✓ 0 (dynamic port mapping)

Memory reservation:

512

CPU:
256

Create task definition.

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Tell us what you think

Create new task definition revision

Task definition configuration

Task definition family: hiring-task-TD2
Revision: 1

▼ Infrastructure requirements

Launch type: Amazon EC2 instances
OS, Architecture, Network mode: Linux/X86_64, bridge

Task size

CPU: 1 vCPU
Memory: 5 GB

Task roles - conditional

Private registry

Port mappings: hiring-task-ttd-8080-tcp, HTTP

Read only root file system

Read only

Resource allocation limits - conditional

CPU: 1, GPU: 1, Memory hard limit: 0.5, Memory soft limit: 0.25

▼ Environment variables - optional

Environment variables: Add environment variable

Add from file

Overview

ARN: arn:aws:ecs:us-east-1:060795932818:task-definition/hiring-task-TD2
Status: ACTIVE
Time created: February 13, 2026, 16:25 (UTC+5:30)
App environment: EC2
Task role: ecs:TaskExecutionRole
Operating system/Architecture: Linux/X86_64
Network mode: bridge

Containers: JSON | Task placement | Volumes (0) | Requires attributes | Tags

Task size

Task CPU: hiring-task-TD

Task memory

Task memory maximum allocation for container memory reservation: hiring-task-TD

▼ Container: hiring-task-TD

Image: labaid509/hiring-app-works
Private registry: turned off
Secrets Manager ARN or name: -
CPU: 256 vCPU
Memory hard/soft limit: 5 GB/1 GB
GPU: 1

Environment and secrets | Network settings | Security and permissions | Lifecycle and dependencies | Monitoring and logging | Runtime configuration | Additional settings

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✅ STEP 8: Create ECS Service (Maintain 6 Tasks Always)

Go to:

ECS → Clusters → your-cluster → Services → Create

Settings:

Launch type: EC2

Service name: hiring-service

Task definition: hiring-task

Desired tasks: ☒ 6

The screenshot shows the AWS Management Console for the Amazon Elastic Container Service (ECS). The breadcrumb navigation at the top indicates the path: Amazon Elastic Container Service > Clusters > hiring-cluster-dev > Services > hiring-task-TD2-service > Update. The left sidebar contains navigation links for Amazon ECR, Repositories, AWS Batch, and other services. The main content area is divided into two sections. The top section, 'Task definition family', shows 'hiring-task-TD2' selected. Below it, 'Task definition revision' is set to '1'. The 'Scheduling strategy' is 'REPLICA'. 'Desired tasks' is set to '6'. 'Availability Zone rebalancing' is checked. 'Health check grace period' is set to '0' seconds. The bottom section, 'Load balancing - optional', shows 'Use load balancing' checked. A 'VPC' is selected, and a 'Load balancer - 1' is configured with 'Load balancer type' as 'Application Load Balancer', 'Listeners' as 'HTTP:8080', and 'Container name:port' as 'hiring-task-TD2:8080'. A 'Target group' is also shown as 'newtargetgroup'.

☒ STEP 9: Configure Load Balancer in Service

In Load balancing section:

Enable:

☒ Application Load Balancer

Select:

- Load balancer: hiring-alb
- Listener: HTTP 80

Create service.

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Tell us what you think

hiring-task-TD-service-pzsgrqh1 deployment is in progress. It takes a few minutes.

hiring-task-TD-service-pzsgrqh1

Status: Active

Tasks (6 Desired): 0 Pending | 0 Running

Task definition: revision hiring-task-TD-1

Deployment status: In progress

Health and metrics | Tasks | Logs | Deployments | Events | Configuration and networking | Service auto scaling | Event history | Tags

Status

Service name: hiring-task-TD-service-pzsgrqh1

Service ARN: arn:aws:ecsus-east-1:060795932818:service/hiring-cluster-dev/hiring-task-TD-service-pzsgrqh1

Deployments current state: 0 Completed tasks

Created at: February 12, 2026, 16:52 (UTC+5:30)

Health check grace period: 0 seconds

▼ Load balancer target health

Container name:port	Target group	Target health	Load balancer	Load balancer type
hiring-tg-TD-80	hiring-tg-ecs Details	No targets	hiring/loadbalancer	Application Load Balancer

✓ STEP 10: Validate Tasks Running

Go to:

ECS → Cluster → Services → hiring-service → Tasks

You must see:

✓ 6 tasks running

If only 2-3 run:

➡ Your EC2 capacity is low.

Increase Auto Scaling desired instances to 3 or change instance type to t3.medium.

Task	Last status	Desired status	Task definition	Health status	Created at	Started by	Started at	Group
08c72870b1d1496b9a24...	Running	Running	hiring-task-TD2:1	Unknown	12 minutes ago	ecs-svc/61900099072...	12 minutes ago	service:hiring-task-TD2...
1444fa2648b4d80c4e0b...	Running	Running	hiring-task-TD2:1	Unknown	38 minutes ago	ecs-svc/61900099072...	36 minutes ago	service:hiring-task-TD2...
d17268daa898481a8155...	Running	Running	hiring-task-TD2:1	Unknown	35 minutes ago	ecs-svc/61900099072...	34 minutes ago	service:hiring-task-TD2...
08b266012da440c38693...	Running	Running	hiring-task-TD2:1	Unknown	12 minutes ago	ecs-svc/61900099072...	12 minutes ago	service:hiring-task-TD2...
da8b24ce528f444ab0a7...	Running	Running	hiring-task-TD2:1	Unknown	38 minutes ago	ecs-svc/61900099072...	34 minutes ago	service:hiring-task-TD2...

✓ STEP 11: Validate Dynamic Port Mapping

Go to:

EC2 → Target Groups → hiring-tg → Targets

You should see registered targets like:

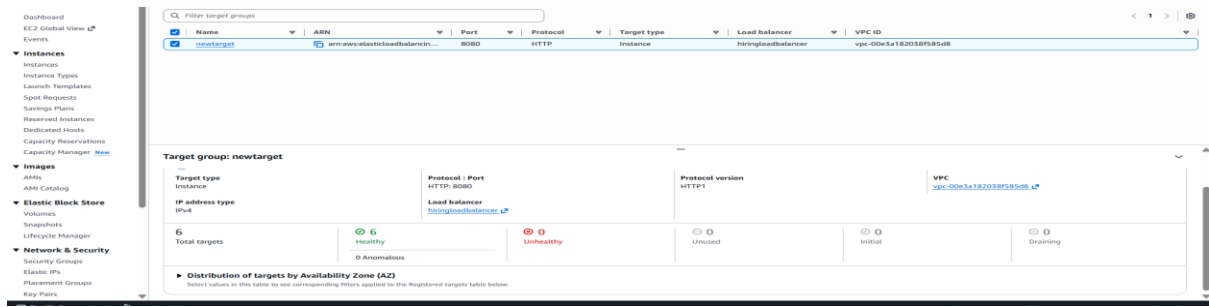
- i-xxxx : 32768
- i-xxxx : 32771

This confirms:

✓ dynamic port mapping working

Task	Last status	Desired status	Task definition	Health status	Created at	Started by	Started at	Container instance	Launch type	Platform version
32b0b07465ee4d9f9fce73aa73a1fc8a	Running	Running	hiring-task-TD2:1	Unknown	1 second ago	ecs-svc/61900099072...	1 second ago	ecs-svc/61900099072...	EC2	Linux
32b0b07465ee4d9f9fce73aa73a1fc8a	Running	Running	hiring-task-TD2:1	Unknown	1 second ago	ecs-svc/61900099072...	1 second ago	ecs-svc/61900099072...	EC2	Linux
32b0b07465ee4d9f9fce73aa73a1fc8a	Running	Running	hiring-task-TD2:1	Unknown	1 second ago	ecs-svc/61900099072...	1 second ago	ecs-svc/61900099072...	EC2	Linux
32b0b07465ee4d9f9fce73aa73a1fc8a	Running	Running	hiring-task-TD2:1	Unknown	1 second ago	ecs-svc/61900099072...	1 second ago	ecs-svc/61900099072...	EC2	Linux
32b0b07465ee4d9f9fce73aa73a1fc8a	Running	Running	hiring-task-TD2:1	Unknown	1 second ago	ecs-svc/61900099072...	1 second ago	ecs-svc/61900099072...	EC2	Linux

→target group



✅ STEP 12: Final Test (ALB DNS)

Go to:

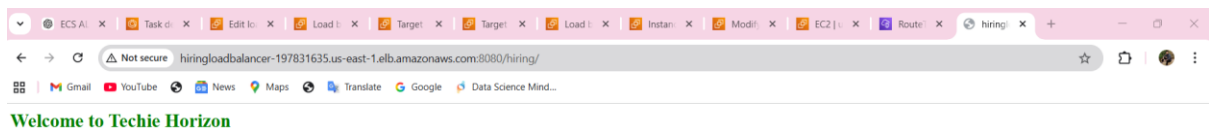
EC2 → Load Balancers → hiring-alb → DNS name

Open DNS in browser:

Example:

<http://hiring-alb-xxxxx.us-east-1.elb.amazonaws.com>

I should see the application running.



✅ FINAL CHECKLIST (Your Task Completed)

- ✅ Cluster EC2 in Multi-AZ
- ✅ ALB working
- ✅ Target group healthy
- ✅ Dynamic port mapping (host port 0)
- ✅ Service running 6 tasks always
- ✅ App accessible through ALB DNS